

CS5487: MACHINE LEARNING: PRINCIPLES AND PRACTICE

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Machine Learning: Principles and Practice

Subject Code

CS - Computer Science

Course Number

5487

Academic Unit

Computer Science (CS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

CS3334 Data Structures

AND

[MA2176 Basic Calculus and Linear Algebra or
MA2170 Linear Algebra & Multi-variable Calculus or
MA2172 Applied Statistics for Sciences & Engineering]

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

The goal of this course is for students to learn the fundamental knowledge needed to design machine learning algorithms.

Machine learning algorithms allow computers to automatically learn to recognize complex patterns from empirical data, such as text and web documents, images, videos, sound, sensor-data, and databases. This course is intended to give a broad overview of machine learning with a focus on fundamental design, derivation, and implementation of machine learning algorithms. At the end of the course, students will have fundamental knowledge needed to design and implement new machine learning algorithms from first principles.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Identify, explain, and derive common machine learning algorithms from first principles mathematically.				
2	Implement machine learning algorithms to solve real-world problems, and demonstrate how the model is learned		x		
3	Analyze the effectiveness of different machine learning algorithms. Compare and assess their relative merits with theoretical principle.		x		
4	Design and create new machine learning algorithms to address algorithmic shortcomings and solve particular problems.			x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Students will engage with selected machine learning algorithms, and their intuition, design principles, and derivations. The algorithms will be illustrated with both toy and real-world examples to motivate the students' understanding. Implementation details will also be discussed.	1, 3	2 hours/week
2	Tutorial	Each week, students will work on problem sets during the tutorial sessions to gain better understanding of the lecture material.	1, 2, 3	1 hour/week
3	Programming Assignment	Students will develop skills implementing selected machine learning algorithms, applying them to small datasets, and interpreting the results. Students can then observe the effectiveness of the algorithm, and evaluate the differences between various algorithms.	2, 3	
4	Theory Assignment	Students will analyze algorithms theoretically, and derive solutions to various machine learning problems.	4	
5	Course Project	Students will design and create a system based on a machine learning algorithm to solve a real-world problem. Students will report their results in a course report and during a poster session held at the end of the semester.	2, 3, 4	

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Assignments	2, 3, 4	30	-	Yes
2	Midterm	1, 3, 4	10	-	No
3	Course Project	2, 3, 4	30	For a student to pass the course, at least 30% of the maximum mark for the examination AND course project must be obtained.	Yes

Continuous Assessment (%)

70

Examination (%)

30

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark for the examination AND course project must be obtained.

Assessment Rubrics (AR)**Assessment Task**

Assignments (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

1.1 ABILITY to APPLY machine learning to problems and INTERPRET the results

1.2 ABILITY to COMPARE the accuracy and efficiency of machine learning algorithms

1.3 ABILITY to DESIGN and DERIVE new machine learning algorithms

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Midterm (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

- 2.1 ABILITY to EXPLAIN machine learning algorithms, and INTERPRET results from machine learning algorithms
- 2.2 ABILITY to EVALUATE, COMPARE, and CONTRAST different machine learning approaches
- 2.3 ABILITY to DESIGN and DERIVE new machine learning algorithms

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Course Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

- 3.1 ABILITY to APPLY machine learning to real-world problems and INTERPRET the results
- 3.2 ABILITY to EVALUATE, COMPARE, and CONTRAST different machine learning algorithms
- 3.3 ABILITY to DESIGN and DERIVE new machine learning algorithms

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

- 4.1 ABILITY to EXPLAIN machine learning algorithms, and INTERPRET results from machine learning algorithms
- 4.2 ABILITY to EVALUATE, COMPARE, and CONTRAST different machine learning approaches

4.3 ABILITY to DESIGN and DERIVE new machine learning algorithms

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Assignments (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

1.1 ABILITY to APPLY machine learning to problems and INTERPRET the results

1.2 ABILITY to COMPARE the accuracy and efficiency of machine learning algorithms

1.3 ABILITY to DESIGN and DERIVE new machine learning algorithms

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Midterm (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

2.1 ABILITY to EXPLAIN machine learning algorithms, and INTERPRET results from machine learning algorithms

2.2 ABILITY to EVALUATE, COMPARE, and CONTRAST different machine learning approaches

2.3 ABILITY to DESIGN and DERIVE new machine learning algorithms

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Course Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

3.1 ABILITY to APPLY machine learning to real-world problems and INTERPRET the results

3.2 ABILITY to EVALUATE, COMPARE, and CONTRAST different machine learning algorithms

3.3 ABILITY to DESIGN and DERIVE new machine learning algorithms

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

4.1 ABILITY to EXPLAIN machine learning algorithms, and INTERPRET results from machine learning algorithms

4.2 ABILITY to EVALUATE, COMPARE, and CONTRAST different machine learning approaches

4.3 ABILITY to DESIGN and DERIVE new machine learning algorithms

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Part III Other Information

Keyword Syllabus

The course will mainly focus on fundamental knowledge of designing and deriving machine learning algorithms from first principles. Topics include statistical learning, data clustering, dimensionality reduction and data visualization,

discriminative classifiers, and regression. Programming assignments will touch the following applications: document analysis, spam detection, document clustering, image segmentation, data visualization, face detection, face recognition.

Syllabus:

- a. Overview of machine learning with real-world examples
- b. Statistical learning
 - i. probability distributions (univariate)
 - ii. parameter estimation (maximum likelihood)
 - iii. Bayes' rule & MAP classifiers
 - iv. Naive Bayes classifier
 - v. multivariate probability distributions
 - vi. Bayesian classifiers
 - vii. exponential family distributions & conjugate priors
- c. Data clustering
 - i. K-means clustering
 - ii. Gaussian mixture models and the EM algorithm
 - iii. KDE and mean-shift clustering
 - iv. Spectral clustering, normalized cuts
- d. Dimensionality reduction and visualization
 - i. subspace methods: unsupervised (PCA, LSA); supervised (LDA)
 - ii. non-linear manifold embedding (LLE, MDS, ISOMAP)
 - iii. Nystrom extension
 - iv. exponential family PCA and pLSA
- e. Discriminative classifiers
 - i. nearest neighbors
 - ii. linear classifiers
 1. Fisher linear discriminant (LDA)
 2. logistic regression, conditional LL
 3. linear SVM
 - iii. non-linear classifiers
 1. kernel SVM
 2. Boosting
 - iv. kernel PCA, kernel K-means, and kernel machines
- f. Regression
 - i. least-squares regression
 - ii. generalized linear models
 - iii. Bayesian regression & Gaussian processes

Reading List

Compulsory Readings

Title	
1	Bishop, "Pattern Recognition and Machine Learning", Springer-Verlag New York, 2006.

Additional Readings

Title	
1	Murphy, "Machine Learning: A Probabilistic Perspective", The MIT Press, 2012.
2	Schölkopf and Smola, "Learning with kernels: support vector machines, regularization, optimization, and Beyond.", The MIT Press, 2001.

3	Rasmussen and Williams, "Gaussian Processes for Machine Learning", The MIT Press, 2006.
4	Duda, Hart, & Stork, "Pattern Classification", Wiley-Interscience; 2 edition, 2000.